

(1.1) Dokažte, že v libovolném komplexním vektorovém prostoru V se skalárním součinem $\langle \cdot, \cdot \rangle$ platí (pro libovolné $u, v \in V$)

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2),$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u - iv\|^2 - \|u + iv\|^2),$$

kde $\operatorname{Re}(x)$ značí reálnou a $\operatorname{Im}(x)$ imaginární část komplexního čísla x . (Pro inspiraci vizte tvrzení 8.32 ve skriptech.)

$$a) \operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2) :$$

$$\|u + v\|^2 = \langle u + v, u + v \rangle = \langle u + v, u \rangle + \langle u + v, v \rangle =$$

$$= \langle u, u \rangle + \langle v, u \rangle + \langle u, v \rangle + \langle v, v \rangle =$$

$$= \|u\|^2 + \|v\|^2 + \langle v, u \rangle + \langle u, v \rangle$$

$$\|u - v\|^2 = \langle u - v, u - v \rangle = \langle u - v, u \rangle + \langle u - v, -v \rangle =$$

$$= \langle u, u \rangle + \langle -v, u \rangle + \langle u, -v \rangle + \langle -v, -v \rangle =$$

$$= \|u\|^2 + (-1)\langle v, u \rangle - \langle u, v \rangle + (-1) \cdot (-1)\langle v, v \rangle =$$

$$= \|u\|^2 - \langle v, u \rangle - \langle u, v \rangle + \|v\|^2$$

$$\|u + v\|^2 - \|u - v\|^2 = \cancel{\|u\|^2} + \cancel{\|v\|^2} + \langle v, u \rangle + \langle u, v \rangle - \cancel{\|u\|^2} - \cancel{\|v\|^2} +$$

$$+ \langle v, u \rangle + \langle u, v \rangle = 2\langle v, u \rangle + 2\langle u, v \rangle =$$

$$= 2(\overline{\langle u, v \rangle} + \langle u, v \rangle)$$

$\forall x \in \mathbb{C}$ platí: $\bar{x} + x = 2 \operatorname{Re}(x)$, jelikož $x = a + bi$,
 $\bar{x} = a - bi$

$$a + bi + a - bi = 2a, \text{ kde } a = \operatorname{Re}(x)$$

$$\|u+v\|^2 - \|u-v\|^2 = 2 \cdot 2 \operatorname{Re}(\langle u, v \rangle) \Rightarrow$$

$$\Rightarrow \operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

b) $2 \operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2) :$

$$\begin{aligned} \|u - iv\|^2 &= \langle u - iv, u - iv \rangle = \langle u - iv, u \rangle + \langle u - iv, -iv \rangle = \\ &= \langle u, u \rangle + \langle -iv, u \rangle + \langle u, -iv \rangle + \langle -iv, -iv \rangle = \end{aligned}$$

$$= \|u\|^2 + (-i) \langle v, u \rangle + (-i) \langle u, v \rangle + (-i) \cdot (-i) \langle v, v \rangle =$$

$$= \|u\|^2 + \|v\|^2 + i \langle v, u \rangle - i \langle u, v \rangle$$

$$\|u + iv\|^2 = \langle u + iv, u + iv \rangle = \langle u + iv, u \rangle + \langle u + iv, iv \rangle =$$

$$= \langle u, u \rangle + \langle iv, u \rangle + \langle u, iv \rangle + \langle iv, iv \rangle =$$

$$= \|u\|^2 + (i) \langle v, u \rangle + i \langle u, v \rangle + i \cdot i \langle v, v \rangle =$$

$$= \|u\|^2 + \|v\|^2 - i \langle v, u \rangle + i \langle u, v \rangle$$

$$\|u - iv\|^2 - \|u + iv\|^2 = \cancel{\|u\|^2} + \cancel{\|v\|^2} + i\langle v, u \rangle - i\langle u, v \rangle - \cancel{\|u\|^2} - \cancel{\|v\|^2} + i\langle v, u \rangle - i\langle u, v \rangle =$$

$$= 2i\langle v, u \rangle - 2i\langle u, v \rangle = 2i(\overline{\langle u, v \rangle} - \langle u, v \rangle) =$$

$$= 2(i(\overline{\langle u, v \rangle} - \langle u, v \rangle))$$

$\forall x \in \mathbb{C} : i(\bar{x} - x) = 2 \operatorname{Im}(x)$, jedoch $x = a + bi, \bar{x} = a - bi$

$$i(a - bi - a - bi) = i(-2bi) = -2b \cdot (i)^2 = -2 \cdot (-1) \cdot b =$$

$$= 2b, \text{ also } b = \operatorname{Im}(x)$$

$$\|u - iv\|^2 - \|u + iv\|^2 = 2 \cdot 2 \operatorname{Im}(\langle u, v \rangle) \Rightarrow$$

$$\Rightarrow \operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2)$$
